Restoring Land through a Robotic End-Effector

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Introduction

How do we design a rover-agnostic robotic end-effector for land-restoration?

Land degradation, caused by **erosion, nutrient depletion, and desertification**, affects **40% of Earth's land**, impacting **3 Billion people** and costing the global economy around **10 trillion USD annually** [1].

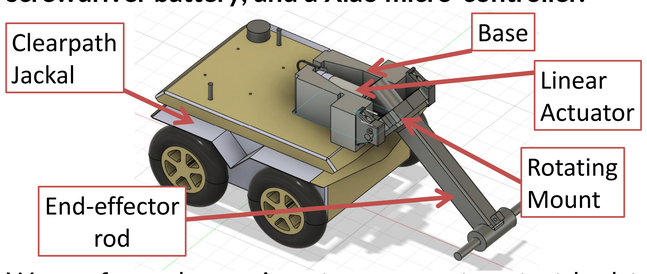


Figure 1: UNCCD/Mwangi Kirubi Small farms in northern Kenya are struggling to grow crops amidst increasingly arid conditions.

Currently, there is **no large-scale solution.** Following Toluwaniase Ademola's '26 approach [2], we propose a **robotic-end effector for berm-digging.** This also fills a gap in the robotics field as the focus is either on precise localized methods or industrial scale digging, **neglecting medium-scale robotic excavation.**

Methods

In the Digital Fabrication Lab (SC) we **re-designed the end-effector and developed a tool-changing mechanism.** The **previous model used the front of the rover** however, most rovers only have mounting holes on the top, so we adapted the design. We used **3D printing, a linear actuator, an aluminum rod, a screwdriver battery, and a Xiao micro-controller.**



We performed experiments on a custom test-bed to emulate a semi-arid compressed soil environment. We **tested the end-effector's endurance, gathered real-time data through ROS2, and determined a method for digging and compressing berms.**

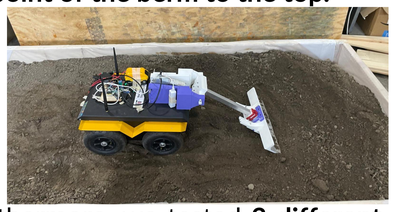
Acknowledgements

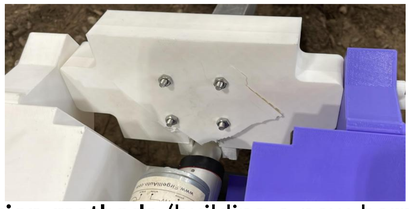
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Results

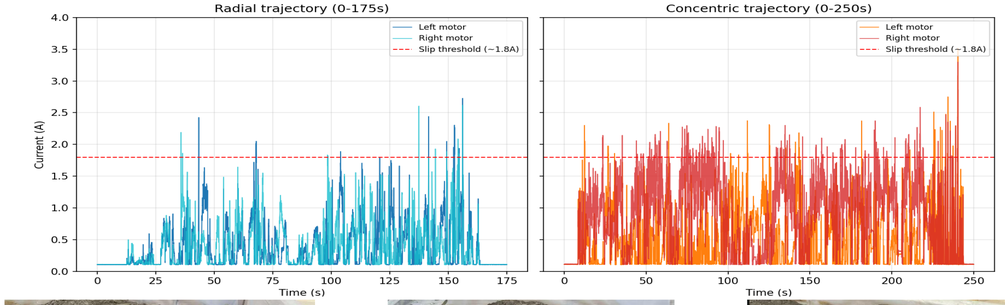
Using the **Jackal to dig semi-circular berms,** we found out the following:

- The **main stress point** is **the back-center of the rotating mount** (we reinforced it)**.**
- **Slippage tends to occur** more often when **turning** (or with **heavy loads** in general).
- When the **Jackal slips**, the **current consumed** by each motor is often >**1.8 Amps**.
- **Several passes** with **5–10-degree plow-angle variations** are an efficient **berm-digging method.** This allowed the Jackal to build taller and wider berms.
- Berms reach **4-4.5 in of height from ground level and ~6.5 inches from the lowest point of the berm to the top.**





Furthermore, we tested **3 different compression methods** (building a new berm for each test) to create **erosion-resistant** berms. The **radial method** went over the berm **following a radial trajectory**. The home point was the center of the circumference of the berm. The **concentric compression method** drove over the berm tracing **expanding concentric circles.** The **intermediate method** performed either pattern in **between passes of the berm digging process** (instead of only once it was finished).

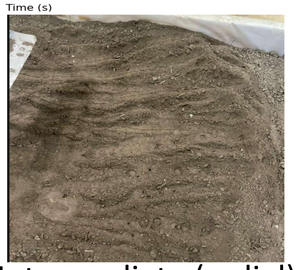




Radial



Concentric



Intermediate (radial)

Concentric trajectories made the motors **surpass the 1.8 Amps limit** and incurred in **slippage more frequently than the radial trajectories**. Furthermore, the **radial method compressed the berm walls to ~0.181 Kg/cm², the concentric ~0.091 Kg/cm², and the intermediate-radial ~0.27-0.54 Kg/cm².** The **radial method left a berm with 4 in. of height to the ground, and 5 in. to its lowest point**, the **concentric with 4.7 in. and 6 in.**, and the **intermediate with 4.5in. and 6.5 in.**

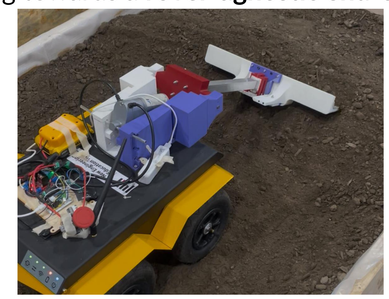
References

[1] *Three billion people globally impacted by land degradation | UN News*. (2024, December 2). https://news.un.org/en/story/2024/12/1157651

[2] Ademola, T., McCarthy, R., Gokongwei, J., Cuevas Vega, D., Kennedy, J. R. M., Bar-Sinai, K. L., & Melenbrink, N. (in press). Assessing the potential utility of low-cost robotic end-effectors for ecological restoration. In *2026 IEEE Global Humanitarian Technology Conference (GHTC)*. IEEE.

Conclusion

- It is **possible to dig significant berms** with a **Clearpath Jackal.**
- This design **mounts exclusively on the top**, building towards a **rover-agnostic end-effector.**



- **Slippage** often occurs when motors consume **>1.8 Amps. Turning & heavy loads** are the main causes.
- The **intermediate-radial method** is the **most efficient** preserving the **berm's dimensions, compressing the walls,** and had **fewer observed slippage events.**
- To **dig a berm**, **several back-and-forth passes** will be needed **alongside varying the plow-angle.**

Next Steps

The team will focus on the following:

- **Try out the end-effector** on **different rovers** to **compare the effectiveness** and experiment with **multi-robot interactions**:
  - Autodesk RMP 440 LE V3
  - Rover Robotics MITI



- Conduct field tests in **Hadley, MA** and **Arizona.**
- **Write a research paper** to publish what was found.
- Integrate a **tool-changing mechanism** and incorporate **reinforcement learning** to build towards a se**mi-autonomous system.**

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